T. Rex Footprints Show How Fast the Giant Walked

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Four **fossilized** footprints are helping scientists step into a dinosaur's shoes. The prints form the first known set of tracks likely left by an adult Tyrannosaurus rex.

A **trackway** is three or more consecutive footprints left by some animal. The one for this T. rex spans 7 meters. The track spacing suggests the ancient predator had been ambling along at about 1.6 meters (5.2 feet) per second, which would be a quick walking pace for us.

A few single adult T. rex footprints have been recorded before. So have **trackways** from what might be T. rex juveniles or another related species, says Peter Falkingham, a **paleontologist** at Liverpool John Moores University in England. 'Finding an adult **trackway** is cool,' he says. 'It's unique. It brings something new.'

The reason: 'An individual [footprint] doesn't tell you a lot,' Falkingham explains. 'A **trackway** tells you more about foot placement as the animal's moving along.' And younger and older dinosaurs likely walked differently, just as human toddlers and adults do.

Kent Hups teaches at Northglenn High School, outside Denver, Colorado. A **paleontologist**, he also works at the Denver Museum of Nature and Science. Hups stumbled on the **trackway** while fossil hunting in North Dakota last year. It stuck up out of the ground along a steep incline in the **Hell Creek Formation**. Sediment had filled in the originally muddy footprints and **fossilized**. It then **eroded** away more slowly than the surrounding rock.

The first print that he found was three-toed and enormous. It measured 80 centimeters (31.5 inches) across. He recalls immediately thinking: 'This has to be a T. rex.'

Slender toe impressions and fossils unearthed in the area signal the **trackway** had been left behind roughly 66.5 million years ago. 'There's always the possibility that we haven't found the skeletons of some species that we don't know about,' Falkingham says. 'But as far as our knowledge goes, T. rex is about the only track-maker that could have made these.'

He and Hups were part of a team that used **stride** length and body size to gauge the dinosaur's speed. Unfortunately, the footprints aren't well-preserved. And estimating speed from tracks is tricky. So the T. rex may have been walking slightly faster or slower than what they reported September 8 in the Journal of Vertebrate Paleontology.

Better tracks could have revealed whether the foot landed flat or toe-to-heel or even how its foot pads supported the dino's weight. More tracks are sure to follow. In fact, a fifth step may be concealed in the hill. But the team is waiting for Mother Nature to wear away the surrounding rock, Hups says. 'I can't wait to get back out there and keep looking for more.'

Word Treasure

trackway -- A trackway is three or more footprints left in a row by an animal.

paleontologist -- A paleontologist is a scientist who studies fossils.

fossilized -- Fossilized means turned into a fossil over a very long time.

stride -- A stride is the length of one step.

eroded -- When rock is eroded, it is slowly worn away.

Hell Creek Formation -- The Hell Creek Formation is a layer of rock in North Dakota where

many dinosaur fossils are found.

Background you need

A fossil is any preserved remains or traces of ancient life, and footprints are a kind called trace fossils.

A paleontologist is a scientist who studies fossils to learn how ancient animals lived.

Tyrannosaurus rex, often called T. rex, was a huge meat-eating dinosaur that lived at the end of the age of dinosaurs.

Scientists can estimate how fast an animal moved by measuring the distance between its footprints and the size of its body.

Break it down

LEAD: Four fossilized footprints form the first known trackway likely left by an adult T. rex.

KEY EVENT: Kent Hups found the tracks while fossil hunting in North Dakota last year.

+- WHO: Kent Hups, a high school teacher and paleontologist, and Peter Falkingham, a paleontologist in England

+- WHAT: Four three-toed footprints spanning 7 meters, the first 80 centimeters across

+- WHEN: The tracks were left roughly 66.5 million years ago

EVIDENCE: Stride length and body size suggest a pace of about 1.6 meters per second, a quick walk for us.

OPEN QUESTION: Could a fifth step be hidden in the hill, and will better tracks reveal more about how the foot moved?

Why it matters

Footprints show how dinosaurs actually moved, not just what their bones looked like, so they bring ancient animals to life.

Test what you remember

1. How many footprints make up the T. rex trackway?

(A) Eighty
(B) Four
(C) Seven
(D) Three

2. How fast did the T. rex appear to be walking?

(A) About 1.6 meters per second
(B) About 6.5 meters per second
(C) About 16 meters per second
(D) About 0.6 meters per second

3. Who discovered the trackway?

(A) A team from the Denver Museum
(B) A student in North Dakota
(C) Peter Falkingham, a teacher in England
(D) Kent Hups, a high school teacher and paleontologist

4. Where were the footprints found?

(A) In the Hell Creek Formation of North Dakota
(B) In a museum in Denver
(C) In the deserts of Montana
(D) In a riverbed in Colorado

5. How wide was the first footprint Hups found?

(A) 80 centimeters across
(B) 8 centimeters across
(C) 800 centimeters across
(D) 31.5 centimeters across

6. Why is an adult trackway special, according to Peter Falkingham?

(A) It is the only fossil ever found in North Dakota
(B) It shows the dinosaur's exact weight
(C) It proves T. rex could run faster than any animal
(D) A trackway shows foot placement as the animal moves, which one footprint cannot

Answer key (try the quiz first!): 1: B 2: A 3: D 4: A 5: A 6: D

Questions to think about

1. Peter Falkingham, paleontologist: Says finding an adult trackway is cool and unique because it shows foot placement as the animal moves.

2. Kent Hups, discoverer: Says he cannot wait to get back out and keep looking for more tracks.

MY THOUGHTS (at least 20 words):